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RESEARCH ARTICLE

FoMoSkinNet: A Dual-Stream Deep Learning Model for Early Detection and Classification of Non-Dermoscopic Skin Lesions via Focal Modulation and Local Feature Integration

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ABSTRACT Skin cancer is a major worldwide health threat, especially melanoma, which has a significantly higher death rate than other types, and early detection significantly improves treatment outcomes. Deep learning architectures, particularly CNN and transformer-based models, have recently emerged as playing an important role in melanoma malignancy detection. However, many existing approaches struggle to generalize effectively when applied to low-quality or non-dermoscopic images exhibiting high visual variability. The primary objective of this work is to enhance skin cancer classification by improving feature representation in non-dermoscopic images. To achieve this, we propose a hybrid approach FoMoSkinNet, that integrates local feature extraction via the Local Feature Network (LFNet) and global feature representation through FocalNet, enabling more accurate skin lesion classification. LFNet combines grouped convolution, channel shuffling, and multi-scale dilated depthwise convolution to capture fine-grained lesion details, whereas the FocalNet module hierarchically aggregates multi-scale contextual information and adaptively modulates feature representations to capture long-range lesion dependencies. The fusion of features from both tracks enhances the extraction of valuable and complementary information. Experimental results demonstrate that FoMoSkinNet achieves 98.85% accuracy, 99.46% precision, 98.23% recall, 98.84% F1-score, and 99.47% specificity on the ISIC2024 dataset, outperforming existing state-of-the-art methods. The comprehensive analysis reflects that the FoMoSkinNet provides an effective and reliable solution for automatic skin cancer detection using non-dermoscopic images.

INDEX TERMS CNN, dilation, focal modulation, gated aggregation, hierarchical contextualization, skin lesion classification.

I. INTRODUCTION

Uncontrolled growth and reproduction of cells in a particular area of the body is known as cancer. It can begin in any body tissue or organ. It is among the most severe diseases that have posed a serious threat to human life recently [1], [2]. Around 14 million instances of cancer and 9.6 million deaths from cancer have been documented worldwide, according to the World Health Organization [3]. When looking at various cancer types, skin cancer appears to have a high risk [4]. The skin, which is the largest organ in the body, serves as a vital shield against damage and infection. Skin cancer can arise when cells in the skin begin to grow aggressively [5]. Skin cancer is defined as an uncontrolled growth of cells in the epidermis, which is the skin's most exterior layer, due to unrepaired destruction of DNA and resulting in mutations. These aberrant skin cells have the ability to spread swiftly to other spots throughout the body, causing cancerous

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lesions. Skin cancer development is influenced by multiple factors. Among these, exposure to ultraviolet (UV) radiation is a major contributor [6]. It induces DNA damage in skin cells, causing mutations that lead to cancer. The skin will often be affected by three types of skin cancer [7]: Basal Cell Carcinoma (BCC), Squamous Cell Carcinoma (SCC) and Melanoma. Basal cells, which reside at the base of the epidermis, are the source of BCC, which is often the most common form. SCC typically grows faster than BCC and is caused by squamous cells that are located near the skin's surface. If untreated, it affect other areas of the body. Melanoma, a very lethal kind of skin cancer, is less common but has steadily increased in incidence rates globally [8]. It occurs due to the growth of melanocytes in response to UV radiation and can appear in various colours, including rose pink, royal purple, azure, and even colourless [9], [10]. According to the possible consequences for skin damage, the majority of researchers believe that melanocytic cancer is malignant because the aberrant melanocytic cells enter the inner layers of the skin, while BCCs are benign [11]. Most skin cancer cases (99%) are basal cell carcinomas and squamous cells, while melanomas make up only 1%. Overall, skin cancer is one of the most widespread and fatal diseases, comprising approximately 1.79% of the global disease strain [12]. In the United States alone, the estimated annual expenditure for treating skin cancer is 8.1$ billion, of which 4.8$ billion is allocated to non-melanoma cases and 3.2$ billion to melanoma. It not only has a high financial cost but also a high human cost, with almost two individuals dying nationwide per hour [13]. The National Cancer Institute estimates that by 2040, 16.4 million people would die from cancer globally, and there would be almost 29.5 million new cases of the disease [4]. Over the previous ten years, different types of skin cancer have become more common. Each year, about 3 million non-melanoma cases and 132,000 melanoma cases are detected globally [14]. It seems to be affecting people under 40 more and more. Women are more likely than men to be diagnosed with skin cancer before the age of fifty. The pattern reverses after age 50, with men being diagnosed at a higher rate than women [15]. Skin cancer can be caused by potential risk factors, including excessive exposure to Ultraviolet (UV) light, frequent sunburns, fair or sensitive skin, a family history of disease, genetic predisposition, a deficient immune system, smoking, and unhealthy lifestyle choices [14]. It is estimated that hundreds of thousands of cases of SCC and BCC are brought on by tanning beds. According to the WHO, users of artificial tanning beds are currently in the highest risk group for developing skin cancer [16]. Despite the fact that melanoma can be aggressive, it is still very treatable if caught early [17]. To ensure effective treatment and improved patient outcomes, early and accurate diagnosis is essential. Non-invasive imaging techniques, including clinical photography, dermoscopy, reflectance confocal microscopy (RCM), optical coherence tomography (OCT), and high frequency ultrasound (HFUS) enable faster and more reliable evaluations while reducing patient discomfort and healthcare costs, as illustrated in Figure 1. The primary objective of this research is to develop a reliable, automated system for skin cancer classification, which is designed to work with real-world, non-dermoscopic clinical images.

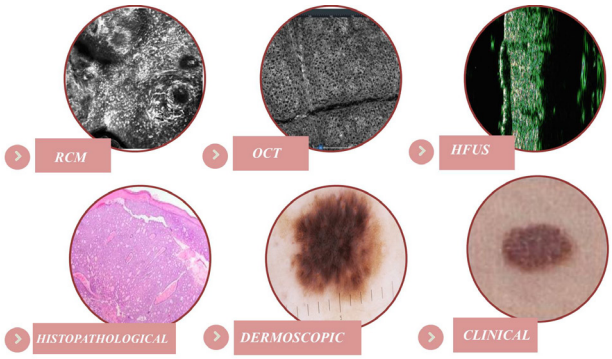


FIGURE 1. Image modalities for skin cancer.

This paper is organized into different sections, each addressing a specific component of the research. These are organized as follows: **Section II** presents a comprehensive review of related literature, discusses computer-aided methods as well as alternative approaches that have been used for relevant issues. **Section III** introduces the dataset used in this study, describing its characteristics and sources. **Section IV** describes the material and methodologies employed, detailing the preprocessing steps, model architecture, and experimental design. **Section V** explains the evaluation measures adopted to assess performance. In **Section VI**, we discuss the results obtained from various experiments, providing analysis, comparisons with existing methods, and insights into the model performance. Finally, **Section VII** concludes the paper by summarizing the key contributions, problems encountered are listed along with recommendations for future work.

II. RELATED WORK

Skin cancer is becoming more common in various populations and has to be diagnosed as soon as possible. Various techniques have been employed for skin cancer prediction, whereas deep learning (DL) has become increasingly popular. Researchers have explored a wide range of methodologies, from conventional machine learning (ML) approaches to advanced DL frameworks. This section provides an overview of the most relevant studies in the field, highlighting their contributions and limitations. Sau et al. [18] proposed an ML-based method for skin cancer classification, using preprocessing, image normalization, and Gray-Level Co-occurrence Matrices (GLCM) for texture feature extraction. They compared SVM, ensemble, and ANN classifiers, with the ANN-SCG model achieving an accuracy of 84.21%. However, their approach relied on a limited dataset and primarily focused on texture features, which may reduce generalizability and overlook other lesion characteristics such

as color and shape. Murugan et al. [19] used texture-based features and a hybrid SVM–Random Forest classifier on 1,000 ISIC images. They applied Gray Level Run Length Matrix (GLRLM), moment invariants, and Gray Level Co-occurrence Matrix (GLCM) for feature extraction, with GLCM, the results (89.31% accuracy, 88.56% sensitivity, 87.81% specificity). A 10-fold cross-validation was used to ensure robustness. However, the limited dataset size may restrict generalizability of their findings. A hybrid approach combining TLBO (Teaching–Learning-Based Optimization) and Extreme Learning Machine (ELM) was proposed by Priyadharshini et al., where the framework included preprocessing, data collection, fuzzy C-Means segmentation, PCA for feature extraction, and ELM-TLBO for classification. The method achieved 93.18% accuracy on a dataset of 300 skin images, SVM (78.11%) and Deep CNN (83.25%) [20]. Gupta et al. proposed a customized CNN. This approach incorporated techniques such as data augmentation, batch normalization, dropout, and dataset balancing, evaluated on the ISIC dataset with 2,239 training and 118 testing images across nine lesion classes. Among three tested models, the final model trained on a balanced dataset achieved the testing accuracy of 89.9% [21]. Bello et al. utilized transfer learning (TL) through pretrained models for skin cancer classification. They combined the ISIC2020 and HAM10000 datasets, totaling 3,297 RGB images, and enhanced the model with additional dense and flattened layers, using LeakyReLU activations to mitigate the vanishing gradient problem. Among several evaluated architectures, the fine-tuned DenseNet121 achieved the accuracy of 87% [22]. DAE-ResNet101, a hybrid deep learning model combining a deep autoencoder for feature reconstruction and a pretrained ResNet101 for classification, was proposed for skin lesion classification on the HAM10000 dataset. The method achieved 96.03% accuracy, 95.40% precision, 96.05% recall, an F1-score of 0.9576, and an AUC of 0.98 [23]. Ali et al. employed a deep CNN model aimed at assisting early detection. The model incorporated preprocessing techniques such as noise reduction, normalization, and augmentation. Evaluated on HAM10000 dataset, it achieved 91.93% accuracy and an AUC of 0.847, offering improved accuracy compared to models, ResNet, VGG-16, DenseNet, and MobileNet [24]. ScATNet, a self-attention-based network for classifying melanocytic lesions in whole slide images (WSIs), was proposed by Wu et al. It first extracted patch-wise embeddings using a CNN, then employed transformers to capture inter-patch and inter-scale correlations. A soft-labeling approach addressed ambiguity in instance-level labels, and max-voting is applied to classify learned representations. The model scored 0.64 accuracy, F1-score of 0.64, sensitivity of 0.88, specificity of 0.79, and ROC-AUC of 0.79 [25]. A Mutual Attention Transformer (MAT) was introduced by Zhou and Luo, which integrated dermoscopic images with textual metadata. The model employed guided and self-attention mechanisms to capture intermodal interactions and preserve contextual information, using Faster R-CNN to extract top-100 regional features from images and GloVe embeddings for textual data. Evaluated on the HAM10000 dataset, MAT achieved 92.55% accuracy and an AUC of 0.98, performing well than state-of-the-art approaches [26]. For automated skin lesion segmentation, Wu et al. proposed FAT-Net, a feature-adaptive transformer network that combined a transformer branch with a traditional encoder-decoder architecture. The dual encoder captures global context via transformer layers and local features via CNNs, while a memory-efficient decoder and feature adaptation module enhance feature fusion and suppress background noise. Evaluated on public datasets (PH2, ISIC 2016–2018), FAT-Net achieved high performance, such as on ISIC 2016: 96.04% accuracy, 85.30% IoU, and 91.59% Dice [27]. A hybrid model that combines CNN and Vision Transformer (ViT) was proposed by Nie et al. High-level semantic information is captured by the transformer by globally modeling the local features that the CNN retrieved. Using focal loss to overcome class imbalance, the model achieved 89.48% accuracy, 82.12% precision, 81.53% recall, 81.10% F1-score, and 0.90 ROC-AUC on the ISIC 2018 dataset, outperforming conventional CNNs (74% accuracy) [6].

Xie et al. presented Swin-SimAM, which combined Swin with a Simple Parameter-free Attention Module (SimAM). SimAM enabled the network to concentrate on important lesion areas without adding parameters. Evaluated on the ISIC-2017 dataset, the model addressed class imbalance using focused loss and achieved an AUC of 0.90 and average precision of 0.74, [28]. Furthermore, Aladhadh et al. suggested a Medical Vision Transformer (MVT) structure. The model used a two-tier approach: the first tier applied data augmentation to increase training samples, and the second tier divided dermoscopic images into patches processed like word embeddings. Evaluated on HAM10000 dataset, model scored accuracy of 96.14%, 96.00% precision, 96.50% recall, and 97.00% F1-score [29]. A ViT-based model by Cirrincione et al. tested for melanoma detection using the ISIC 2017 dataset. By capturing long-range spatial relationships, the model achieved 94.8% accuracy, 92.8% sensitivity, 96.7% specificity, and 0.948 AUROC. However, training dynamics indicated potential overfitting, highlighting the need for regularization or data augmentation [30]. Ozdemir and Pacal presented a hybrid DL model that combined ConvNeXtV2 blocks with focal self-attention. The model outperformed SwinV2-Base (+3.3% accuracy) and ResNet50 (+10.8% accuracy) with accuracy of 93.60%, 91.69% precision, 90.05% recall, and 90.73% F1-score using sophisticated augmentation and TL to solve class imbalance [31]. However, it struggled with visually similar classes such as SCC and actinic keratosis, potentially leading to missed positive cases.

Despite these advancements, there are also a number of persistent issues. Many studies use small datasets that are not representative of real-world situations, particularly for

rare lesion types. The majority of models are trained on high-quality dermoscopic images, while low-quality or non-dermoscopic images are frequently found in actual clinical situations. Some works simply concentrate on accuracy, ignoring sensitivity and specificity, which are important for medical diagnosis. Some works lose important data by undersampling innocuous images. Even though a number of current models claim to have high overall accuracy, they frequently fall low in identifying specific crucial classes. Many current models only adequately address either local or global features. The hybrid models that combine local feature extraction have been the subject of many studies, but they entail substantial processing effort. To address these limitations, we propose FoMoSkinNet, a hybrid deep learning framework designed to effectively integrate local texture features with global contextual information, while ensuring strong generalization to real-world and low-quality skin lesion datasets.

A. RESEARCH GAP

This research seeks to address the following existing research gaps within the domain of skin cancer classification.

1) The majority of previous research has mostly focused on local feature extraction, paying not much consideration to the contextual interactions between faraway areas in lesion images. Therefore, for more accurate skin lesion analysis, methods that can jointly model both local and global representations are desperately needed.
2) Despite being extensively used for diagnosing skin cancer, pre-trained deep learning models still have a limited capacity to generalize across a variety of lesion types. These models frequently fall short in accurately differentiating subtle differences between benign and malignant tumors. To improve classification accuracy, pre-trained architectures must be further optimized and fine-tuned
3) Although Convolutional Neural Networks have shown notable performance in visual identification, their limited receptive field hinders their capacity to capture spatial context and long-range dependencies. This restriction becomes crucial when small visual clues are dispersed throughout the lesion, requiring a more thorough contextual interpretation.
4) Most of the previous research has focused on dermoscopic images, which are obtained under carefully regulated imaging circumstances. However, the lighting, texture, and background vary more in real-world clinical or non-dermoscopic photographs. This underscores the need for research that successfully manages such domain differences, as models trained exclusively on dermoscopic datasets may not translate well to non-dermoscopic environments.

B. RESEARCH CONTRIBUTIONS

The key contributions are:

1) This research introduces the FoMoSkinNet, a novel architecture for skin cancer classification using low-resolution, non-dermoscopic lesion images. In order to help the model deal with significant class imbalances in the dataset, we have employed augmentation and TL as well.
2) The proposed method extracts both local and global information by integrating features from two different tracks. As a result, the model can more effectively contextualize the learned characteristics and improve classification accuracy without sacrificing crucial information.
3) The proposed model contains a FocalNet track, which effectively captures long-range dependencies and global structural information in lesion images.
4) The proposed model contains an LFNet track, which is CNN based, designed to capture fine-grained spatial patterns from lesion images.

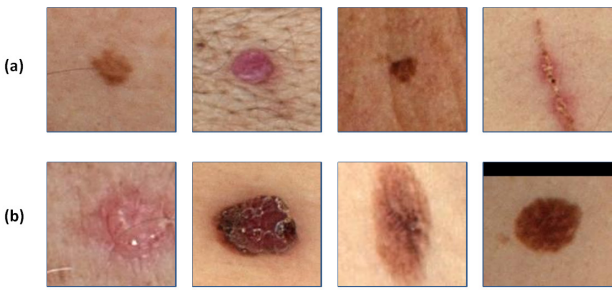


FIGURE 2. Sample images from dataset. (a) Benign lesions (b) Malignant lesions.

III. DATASET

An extensive collection of annotated training and test images, usually a minimum of several thousand, is required to build AI-based image analysis techniques, especially those that use deep CNNs. We have used the ISIC 2024 dataset for the evaluation of our proposed FoMoSkinNet predictor. In particular, SLICE-3D (Skin Lesion Image Crops Extracted from 3D) subset is considered among the most clinically relevant benchmarks for automated skin lesion analysis. It was obtained through the ISIC 2024 Challenge [32]. The ISIC 2024 dataset uses a binary target variable, where 0 represents benign lesions and 1 represents malignant lesions. The dataset includes all lesions from a group of thousands of patients examined in nine institutions across three continents between 2015 and 2024. The standardized photos in this collection were taken with a uniform field of view (FOV) and device model. Patient-captured photos differ significantly in lighting and FOV, yet both image resolution and detail content are similar to those taken with smartphone cameras. This captures real-world differences in lesion anatomy, patient demographics, and imaging settings,

increasing the dataset's usefulness for the development of DL models. Unlike typical dermoscopic imagery, these images were acquired from 3D total body photography (TBP), which provides a clinically realistic portrayal of lesions as shown in Figure 2. Comprehensive metadata, such as a unique image identifier, patient demographics, the lesion's anatomical location, and lesion-specific attributes, are also included in the dataset. Each photograph in the collection is saved in JPEG format and cropped to a 15 by 15 mm area, with an average size of 133 × 133 pixels.

IV. PROPOSED METHOD

In this section, we present FoMoSkinNet, a novel architecture for skin cancer classification using low-resolution, non-dermoscopic lesion images. The FoMoSkinNet extracts both local and global information by integrating features from two different tracks. As a result, the model can more effectively contextualize the learned characteristics and improve classification accuracy without sacrificing crucial information. The overall architecture of FoMoSkinNet is shown in Figure 3.

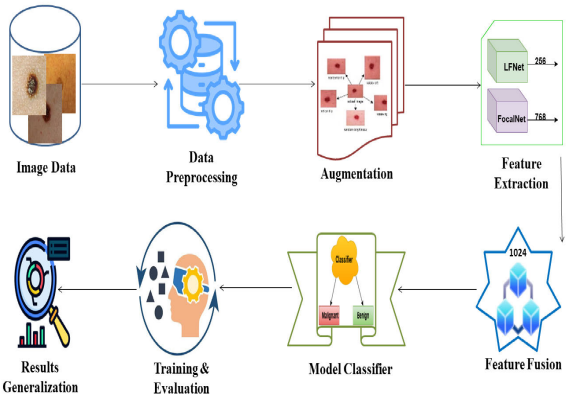


FIGURE 3. Abstract overview of the proposed work.

The workflow is divided into the following subsections.

A. DATA PREPROCESSING

To get raw data ready for model training and evaluation, a number of processes must be applied. The dataset in this study has undergone various processing methods. These procedures are carried out following comprehension of the several salient features of the domain problem. The steps are explained below.

1) IMAGE RESIZING

One of the most basic preprocessing stages in computer vision tasks is image resizing. It guarantees that every skin lesion image has the same shape, which is necessary for the majority of DL models. The dataset we have used contains images of varying sizes. Therefore, these images must be resized to a fixed size that is constant throughout the entire dataset. In this preprocessing step, we have resized lesion images to 224 × 224 × 3 pixels. It is the most common image size that aligns with the model's expected input size [33].

2) IMAGE NORMALIZATION

Image normalization is another preprocessing step that entails modifying an image's pixel intensity values to fall within a specific numerical range. Pixel values in a raw image usually fall between 0 and 255 for each channel. However, because of the significant input scale discrepancies, sending these raw numbers straight into a neural network may result in ineffective learning. After resizing the images, we have scaled the image pixel values to a range of 0 to 1, followed by channel-wise standardization to normalize the skin lesion images [34]. For that, we have used ImageNet mean and standard deviation values, which are frequently used in pretrained models. For each of the Red, Green, and Blue channels, the mean pixel value is subtracted, and the result is divided by the standard deviation. Mathematically, it can be calculated as:

$$\text{pixel}_{\text{norm}} = \frac{\text{pixel} - \mu}{\sigma} \tag{1}$$

where;

- $\text{pixel}_{\text{norm}}$ is normalized pixel,
- pixel is the actual value of the target pixel being normalized,
- μ is mean, and
- σ is standard deviation.

The values used are:

$\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$

B. DATA AUGMENTATION

Data augmentation is the process of applying various label-preserving modifications to a dataset to artificially increase its size and diversity. Usually, there exists a strong link between a model's representational capacity and the volume of training data it represents [35]. Furthermore, a popular and crucial problem in medical image analysis, especially in classification tasks, is class imbalance, which occurs when one class greatly outnumbers the others. This imbalance is caused by the rarity of some medical and health conditions. Ignoring this mismatch can result in a biased learning process where the model favors the majority class. This leads to poor generalization performance and overfitting on unseen data. These problems are tackled by using augmentation, which enhances the model's ability to generalize throughout all classes, particularly the minority classes. With the intention to proceed with our work, we have applied random flipping, rotation, random cropping, contrast enhancement, random shift and scale, and color adjustment [36], shown in Figure 4. Flipping is a commonly used geometric transformation in image augmentation. We apply horizontal and vertical flipping so that our FoMoSkinNet can handle variations in orientation. Rotation alters the spatial arrangement of pixel coordinates and helps the model learn that the lesion region may appear in different orientations, so it is applied with a

limit of ±30 degrees. Color jittering helps the model learn variations in image brightness, contrast, saturation, and hue to improve robustness to lighting conditions, and is applied with brightness, contrast, and saturation factors of 0.3 and a hue factor of 0.1. Random cropping is used to handle spatial variations in skin lesion images. Random shift and scale help simulate real-world situations where lesions or anatomical features may not be centered, and are applied with a shift limit of 0.1, scale limit of 0.1, and border mode set to replicate to fill empty areas. We have also used Contrast Limited Adaptive Histogram Equalization (CLAHE) with a clip limit of 1.5, which enhances skin lesion image with low contrast.

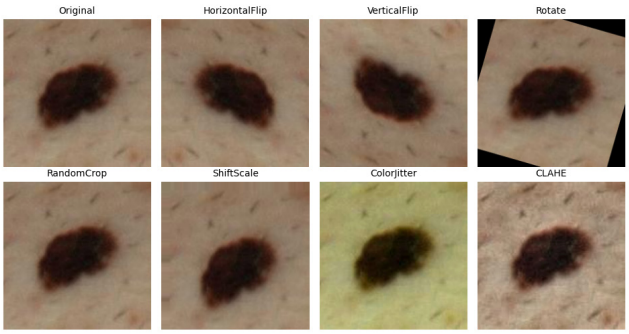


FIGURE 4. Augmented skin lesion samples.

C. DATA SPLITTING

Data splitting is a strategy used for assessing and validating models in ML. It involves dividing datasets into sets for testing and training. The training set contains an extensive portion of the data with labeled samples. It enables the model to learn about the patterns. Conversely, the testing set is used to evaluate model performance on unseen inputs. It is made up of entirely distinct data that model has never seen before [37]. The original ISIC 2024 SLICE-3D dataset is highly imbalanced, with benign cases far outnumbering malignant cases. To avoid classifier bias and ensure computational feasibility, we have used a balanced dataset consisting of 30,000 dermoscopic images, with 15,000 benign and 15,000 malignant samples. This dataset was divided into 80% training and 20% testing, with 10% of the training set further used for validation. The detailed class-wise distribution is summarized in Table 1, while Figure 5 visually illustrates the distribution of benign and malignant cases across the training, validation, and testing splits.

TABLE 1. Dataset distribution and splitting.

Data Split	Benign	Malignant	Total
Training	10800	10800	21600
Validation	1200	1200	2400
Testing	3000	3000	6000
Total	15000	15000	30000

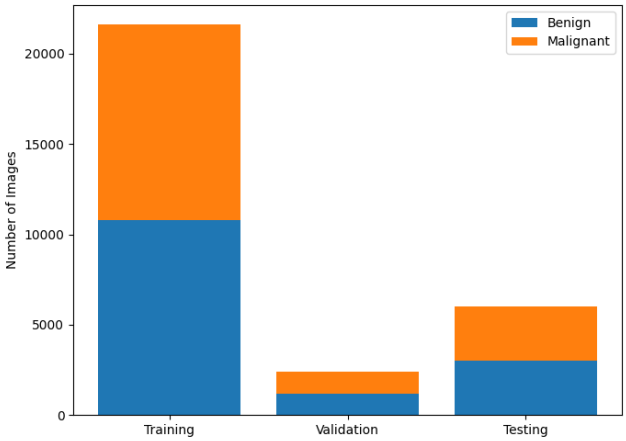


FIGURE 5. Class distribution across dataset splits.

D. FEATURE EXTRACTION

Feature extraction is a technique for extracting significant and valuable features from raw input to enhance the performance of ML models. An image is a combination of pixels. Each pixel is considered a feature of the image. These images have some patterns or regions of interest that help model to classify different images. There are two methods to find a specific pattern or region in skin lesion images. These methods are handcrafted feature extraction and automatic feature extraction using models. Handcrafted method used to extract texture, shape, and edge-related information like Oriented Gradients and Local Binary Patterns Histogram [38]. However, these methods are particularly helpful with smaller-scale or simpler datasets and are dependent on domain knowledge. But as deep learning has progressed, automatic feature extraction with models like CNNs and transformers has improved. Without any manual design required, these models automatically extract hierarchical representations from data, capturing both high-level and fine-grained information [39]. In this work, two different architectures are used for feature extraction: a transformer-based model called Focal Modulation Network (FocalNet) [40] and a novel customized CNN known as LFNet, as shown in Figure 6. The purpose of the novel designed LFNet is to extract low-level, fine-grained features from skin lesion photos, including edges, textures, and localized patterns. This is crucial for identifying minute variations in appearance that conventional models might miss. On the other hand, global contextual and high-level semantic characteristics are extracted using FocalNet. This combination is particularly effective for non-dermoscopic images, which often contain clutter, illumination variations, and weak boundaries; therefore, incorporating both local texture and global context enhances feature representation and overall model performance.

1) FOCAL MODULATION NETWORK (FOCALNET)

In order to encode global structural information and long-range relationships in lesion images, we have

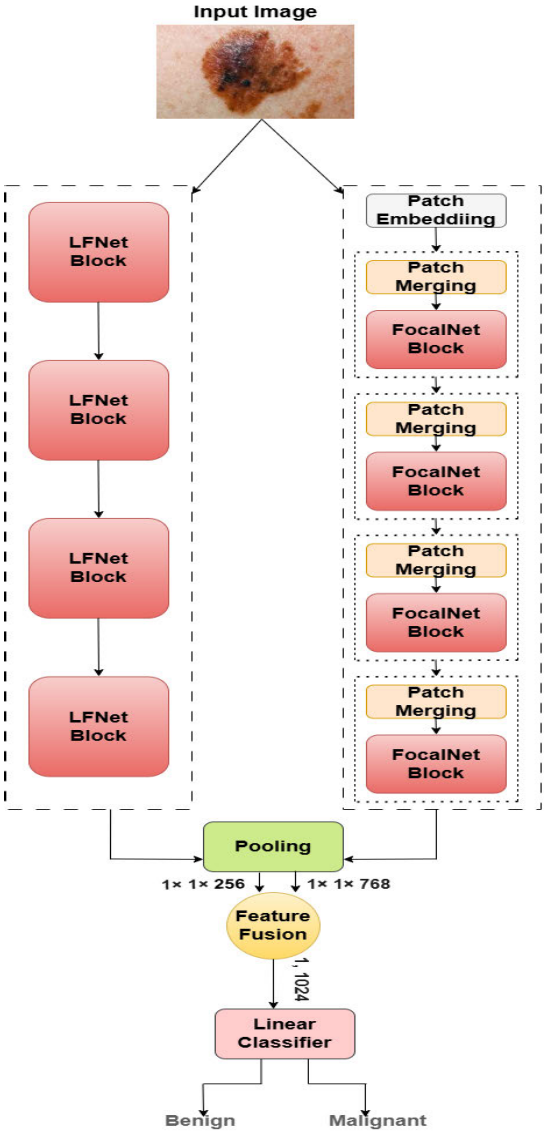


FIGURE 6. Detailed architecture of FoMoSkinNet.

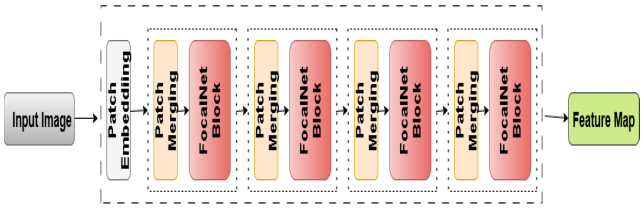


FIGURE 7. Overview of focal modulation network pipeline.

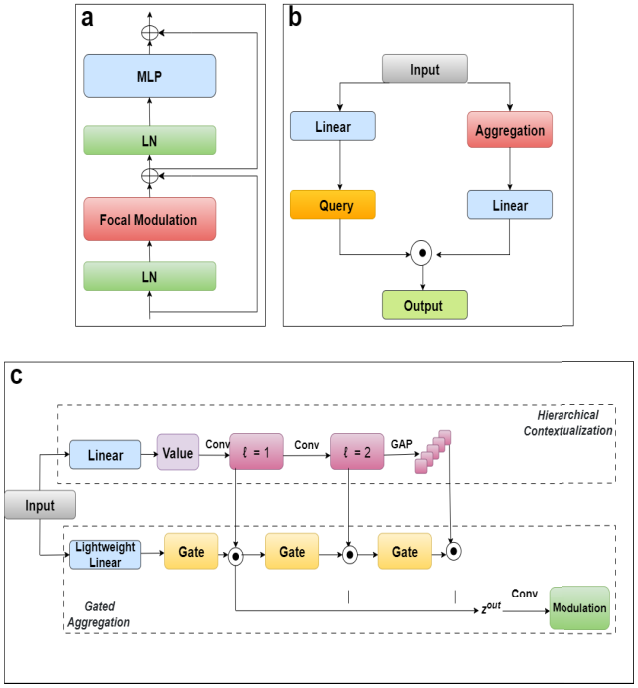


FIGURE 8. Structure of main FocalNetBlock and its parts: (a) FocalNetBlock, (b) Modulation unit and (c) Context aggregation.

incorporated the Focal Modulation Network (FocalNet) as one of the two primary feature extraction. FocalNet uses a focal modulation method in place of traditional self-attention, enabling each token to efficiently and adaptively collect and integrate multi-level contextual information. The FocalNet branch is initialized with pretrained weights to exploit generic visual representations. FocalNet follows a 4-stage hierarchical structure. The network begins with a patch embedding layer, where the lesion image is split into non-overlapping patches of size 4×4, and these patches are linearly projected into an embedding space, forming the initial feature map. This result is passed into **Stage 1**, which begins with a patch merging layer to reduce the resolution and increase the channel depth, followed by 2 FocalNetBlock layers, producing an output of size $56 \times 56 \times C$, where C denotes the image channel. **Stage 2** begins with another patch merging step that downsamples to 28×28 and doubles the channels to $2C$ (i.e., 192), followed by 2 FocalNetBlock layers to capture mid-level semantic information. **Stage 3** further reduces the resolution to 14×14 while increasing the channels to $4C$ (e.g., 384). This stage has 6 FocalNetBlock layers, which focus on capturing high-level global patterns. **Stage 4** performs one more patch merging operation, maintaining the 7×7 spatial shape and increasing the channels to $8C$ (768), followed by 2 FocalNetBlock layers. After all stages, the final output feature map is of size $7 \times 7 \times 768$ as shown in Figure 7.

FocalNetBlock introduces a focal modulation mechanism, which first aggregates context hierarchically and then modulates each token based on that context. It has three main parts: hierarchical contextualization, gated aggregation and element-wise modulation. Each of them is explained below and shown in Figure 8.

In hierarchical contextualization, we extract contexts at various granularities from local to global ranges. This is done by using stack depth-wise convolutions such as 3×3, 5×5, or 7×7 kernels. At the beginning of this block, the input feature map, which is $X \in \mathbb{R}^{H \times W \times C}$ first passed through

a linear layer to convert it into a new feature space: $Z_0 = f_z(X)$. After this, multiple depth-wise convolution layers are applied one after the other. Each one makes a larger portion of the skin lesion image visible to the model, The contextual representation at the ℓ-th level is computed recursively as:

$$Z^{\ell} = f_a^{\ell}(Z^{\ell-1}) \triangleq \text{GeLU}(\text{DWConv}(Z^{\ell-1})) \in \mathbb{R}^{H\times W\times C} \tag{2}$$

where ℓ denotes the index of the current contextualization level, i.e., $\ell \in \{1, 2, \ldots, L\}$, while L represents the total number of hierarchical levels formed by stacking depth-wise convolution layers. The function f_a^{ℓ} denotes the contextualization operation at the ℓ-th level and is implemented using a depth-wise convolution (DWConv) with kernel size k_ℓ, followed by a GeLU activation function. This recursive formulation creates a hierarchy of feature maps $\{Z_1, Z_2, \ldots, Z_L\}$, where deeper levels progressively integrate broader spatial information. Accordingly, (2) generates L levels of feature maps in the hierarchy. The effective receptive field at the ℓ-th level is computed as:

$$r^{\ell} = 1 + \sum_{i=1}^{\ell}(k^{\ell} - 1) \tag{3}$$

where k^{ℓ} is the kernel size used at level ℓ. This allows small convolutional kernels (e.g., 3×3) to approximate large spatial coverage through stacking. To capture holistic scene understanding, we further apply global average pooling to final-level feature map: $Z_{L+1} = \text{AvgPool}(Z_L)$. This yields a total of $L + 1$ multi-scale context features which together capture dependencies with short and long ranges.

In the FocalNet block, the gated aggregation method follows the hierarchical contextualization block, which generates multiple levels of context features. Its goal is to adaptively combine these multi-scale feature maps, allowing model to analyze the relative significance of each level of context based on image content. It might be necessary for each pixel or token in the image to concentrate more on local, mid-range, or global information. A tiny lesion, for instance, could only require local features, whereas a larger or more dispersed lesion might necessitate a more global knowledge. To make this decision, the model learns a set of gating weights for each level of context. The feature map input is placed through a simple, learnable process to obtain these weights, denoted as $G = f_g(X)$, $G \in \mathbb{R}^{H\times W\times(L+1)}$. Here, G contains $L + 1$ values at each spatial location, one for each level of context. These values are used to weight the multi-level feature maps $\{Z_1, Z_2, \ldots, Z_L, Z_{L+1}\}$, which were generated earlier from hierarchical contextualization. The final aggregation is computed by element-wise multiplying the feature map of each level with the associated gate and adding up all the levels, as shown:

$$Z_{\text{out}} = \sum_{\ell=1}^{L+1} G_\ell \odot Z_\ell \tag{4}$$

where Z_ℓ is feature map at level ℓ, G_ℓ is gate weight at that level, and $\odot$ represents element-wise multiplication. After aggregating multi-level context features through gated aggregation, the next step in the FocalNet block is element-wise modulation. This part is responsible for immediately incorporating the contextual data into every token, assisting the network in improving its predictions based on channel-level and spatial features. Up to this point, all operations are focused on the spatial dimension. However, to allow communication across feature channels, a linear layer is applied to the aggregated context map. This transformation is denoted as $h(\cdot)$, as shown in (5), and the output is called the modulator map.

$$M = h(Z_{\text{out}}), \quad M \in \mathbb{R}^{H\times W\times C} \tag{5}$$

Now, for every patch, this modulator holds context-aware data that is organized to represent its position and meaning across channels. At last, each patch undergoes the focal modulation operation. In order to accomplish this, the query vector $q(x_i)$ is multiplied element-wise by the matching modulator vector as shown:

$$y_i = q(x_i) \odot h\left(\sum_{\ell=1}^{L+1} g_{i\ell} \cdot z_{i\ell}\right) \tag{6}$$

where $q(x_i)$ is the query projection of input token x_i, $g_{i\ell}$ and $z_{i\ell}$ are gating value and context feature at location i of g_ℓ and z_ℓ, $h(\cdot)$ is a linear layer that mixes channel information, and $\odot$ represents element-wise multiplication. As a result, we get a modulated token as output y_i, which carries enriched information drawn from different levels of the skin lesion image, making it more effective for complex tasks. Table 2 summarizes the key parameters and their corresponding dimensions used in the FocalNet formulation.

TABLE 2. FocalNet notation summary.

Notation	Description	Dimension
X	Input feature map	H×W×C
Z_0	Linearly projected feature map	H×W×C
Z^{ℓ}	Context feature map at level ℓ	H×W×C
L	Number of context levels	Scalar
ℓ	Index of the current context level	Scalar
k^{ℓ}	Kernel size at level ℓ	Scalar
r_ℓ	Effective receptive field at level ℓ	Scalar
Z_{L+1}	Global context feature (AvgPool)	1×1×C
G	Gating weights for context aggregation	H×W×(L+1)
G_ℓ	Gate value at level ℓ	H×W
Z_{out}	Aggregated context feature	H×W×C
M	Modulator map	H×W×C
x_i	Input token at spatial position i	C
$q(x_i)$	Query projection of token x_i	C
y_i	Output modulated token	C

2) LOCAL FEATURE NETWORK (LFNet)

A novel Local Feature Network is another important component of the proposed architecture, which is designed

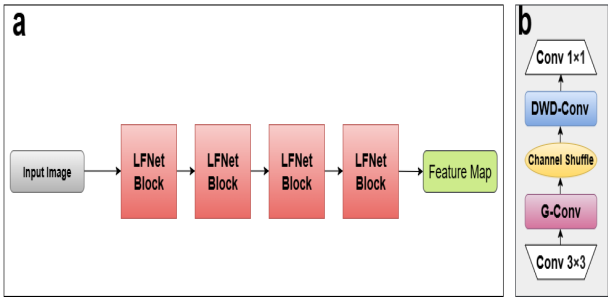


FIGURE 9. (a) Overview of LFNet pipeline, and (b) Structure of a LFNet block.

to capture fine-grained spatial patterns from lesion images, shown in Figure 9(a).

In this block, we employ a combination of standard, grouped, and dilated depthwise convolutions together with channel shuffling [41]. These components, illustrated in Figure 9(b), are designed to capture diverse spatial and contextual information while maintaining computational efficiency, and are described in detail below. First of all the lesion image $X \in \mathbb{R}^{B\times C\times H\times W}$ is passed through standard 3 $\times$ 3 convolutional layer. The purpose of this operation is to increase the channel dimensionality, allowing the network to represent richer low-level features such as edges and textures, which serve as the foundation for subsequent higher-level feature extraction. It can be mathematically represented as:

$$F_1 = X * K_{3\times3} \tag{7}$$

where X is input, $*$ denotes convolution operation and $K_{3\times3}$ represents the 3 $\times$ 3 convolution kernel. The feature map is then subjected to a grouped convolution. This method promotes group-specific feature specialization and increases computing efficiency by splitting the input channels into smaller groups for independent processing. It is formulated as:

$$F_2 = F_1 *_g K_{3\times3} \tag{8}$$

where $*_g$ indicates grouped convolution, and g is the number of groups, where we set $g = 4$. Next, we have applied the shuffle operation. The purpose of a channel shuffle operation is to help groups communicate with each other. It prevents isolated group learning by rearranging the feature channels among groups, improving feature fusion. It is a permutation operation defined as:

$$F_3 = \text{Shuffle}(F_2) \tag{9}$$

After that, the shuffled output is then passed through a series of depthwise dilated convolutions that are intended to capture contextual features at multiple scales. These convolutions work independently on every channel, gradually expanding their receptive field by dilation. It can be represented mathematically as:

$$F_4 = F_3 *_{\text{dw},d} K_d \tag{10}$$

where $*_{\text{dw},d}$ is the depthwise convolution with dilation d, and $K^{(d)}$ is the depthwise kernel corresponding to dilation rate $d \in \{1, 2, 3\}$. Following extraction of multi-scale features, these feature maps are projected to the intended output dimension using a 1 $\times$ 1 pointwise convolution C_{out}. This helps integrate information across channels, as shown:

$$F_5 = F_4 * K_{1\times1} \tag{11}$$

Batch normalization and ReLU activations are applied throughout the block for stable training and improved non-linearity. Finally, we get refined feature map $F_{\text{out}} \in \mathbb{R}^{B\times C_{\text{out}}\times H'\times W'}$ which provides local texture-aware representations, making the proposed model more effective in acquiring each of fine-grained lesion details and broader contextual structures. The notations and dimensions used in the LFNet formulation are summarized in Table 3.

TABLE 3. LFNet notation summary.

Notation	Description	Dimension
X	Input feature map	H×W×C
$K_{3\times3}$	convolution kernel	3×3
F_1	Feature map after standard convolution	H×W×C_1
g	Number of groups	Scalar (g=4)
F_2	Feature map after grouped convolution	H×W×C_1
F_3	Feature map after channel shuffle	H×W×C_1
K_{d}	Depthwise dilated convolution kernel	$d \in \{1, 2, 3\}$
F_4	depthwise feature map	H×W×C_1
$K_{1\times1}$	Pointwise convolution kernel	1×1
F_{out}	Final output feature map	$H' \times W' \times C_{\text{out}}$

E. FEATURE FUSION

In DL, feature fusion is the process of merging features from many pathways to provide a more discriminative and informative representation for tasks like detection or classification [42]. It combines multiple feature vectors into one resultant vector. The diverse sets of features of different models increase the probability of those missing features that are important for skin cancer diagnosis. This approach works especially well when different models capture complementary data; for example, one model may be more adept at catching local patterns while another captures global context. In the proposed method, we combine features of two architectures, which are LFNet and FocalNet, to improve the diversity of pertinent features with concern for skin cancer.

F. CLASSIFICATION

After the feature extraction and fusion phases, the classification head receives the final fused feature map, which includes local texture-aware features from the LFNet block as well as global representations from FocalNet. When we categorize medical images such as skin lesions, we divide them into diagnostic categories (e.g. benign or malignant) according to their traits. The goal of this module is to map the rich, high-dimensional feature vector into a class prediction score that reflects the likelihood of malignancy. In the

FoMoSkinNet, we have a top layer, which is a classification layer. It consists of a series of fully connected operations that include several optimization-related components. First, the fused lesion feature map is flattened into a 1D vector. This is followed by a Layer Normalization (LN) operation to stabilize the distribution of feature values across the batch, thus improving convergence. Fully connected layers are employed to translate learned lesion representations into a meaningful diagnostic output. To avoid overfitting, a dropout layer with a rate of 0.5 is applied next. For optimization, the model employs the AdamW optimizer, which is an Adam variant that decouples weight decay from gradient updates, alongside utilizing Binary Cross-Entropy loss. This helps improve generalization by preventing the weights from growing too large and reducing overconfidence in predictions. To further prevent overfitting, early stopping is applied. At the end of every epoch, the model tracks the validation loss and halts training if no improvement is observed.

V. EVALUATION MEASURES

For assessing the effectiveness of the FoMoSkinNet, we employ numerous evaluation metrics, including accuracy, F1-score, specificity, precision, sensitivity, receiving operating characteristic curve (ROC), and area under ROC (AUC), and confusion matrix.

A. ACCURACY

The degree to which a model's predictions resemble actual values is called accuracy. It is obtained by dividing number of correctly identified samples by the total number of samples [43]. It is calculated using:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{12}$$

where TP and TN denote the number of samples is correctly predicted. FP and FN represent the number of samples that are predicted incorrectly.

B. SENSITIVITY

The sensitivity of the model assesses the total number of real positive cases that the model accurately detects [44]. It is calculated as:

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

C. SPECIFICITY

A model's specificity is determined by how many true negative samples it accurately detects [27]. It is calculated as:

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{14}$$

D. PRECISION

Precision is stated as the likelihood of accurately identifying positive samples among all samples that the model predicts as positive [45]. It represents the number of cases predicted as malignant while they are actually benign. It is calculated as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

E. F1-SCORE

The F1-score provides a fair assessment of both precision and recall by taking the harmonic mean of both [46]. It considers the incorrect predictions when calculating the F1 measure. It is calculated using:

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{16}$$

VI. EXPERIMENTAL RESULTS

To assess the performance of FoMoSkinNet, extensive experiments are performed under a well-defined environmental setup. The FoMoSkinNet's efficiency and robustness are further validated by thorough quantitative and qualitative analysis.

A. SYSTEM CONFIGURATION

All the experiments are performed on a computer system with an Intel (R) Core i9-11900Kf 3.50GHz CPU and 64 GB installed RAM. It also has an NVIDIA 12GB graphical processing unit (GPU). Furthermore, Python is selected for its simplicity, and the PyTorch framework is used to design, train, and analyze models. For the Integrated Development Environment (IDE), we have chosen Visual Studio Code because it offers an interface that is simple to utilize for writing code, displaying data, and tracking training progress. The complete list of experimental tools and environments utilized is summarized in Table 4.

TABLE 4. Hardware and software specifications.

Parameter	Configuration
CPU	Intel(R) Core i9-11900
GPU	NVIDIA 12GB GPU
RAM	64GB
Operating System	Windows 10 (64-bit)
Python	3.12.9
PyTorch	2.5.1
CUDA	12.4
IDE	Visual Studio Code, Jupyter Notebook

B. HYPERPARAMETERS

The performance of DL models is highly affected by the choice of hyperparameters. Training the model involves learning the optimal weights that maps the features of each input instance in the dataset to its corresponding output label. To ensure that this mapping is performed effectively, we have included different parameters such as the loss function, optimizer, learning rate(LR), and weight decay(regularization) that help the training process and improve the generalization ability of FoMoSkinNet. The loss

function guides the model in learning the correct mapping. Optimizer updates the weights of the model based on the loss value and determines how the model learns during training. The regularization parameter adds a penalty to large weight values to prevent overfitting. Another parameter in DL is the learning rate (LR). It controls the size of steps taken during weight updates. LR's value needs to be fixed in such a way that training reaches the global optima. If we increase the value of LR, the step size of convergence also increases and vice versa. So the large value may cause overshooting the global optima. In this study, we have used AdamW as an optimizer and binary cross-entropy as a loss function. In experiments, we have used various parameters with different values and fixed each parameter's value where it performs better than other values. Table 5 presents the selected hyperparameters along with their values used in our experiments.

TABLE 5. Hyperparameters and their values.

Parameter	Value
Image size	224 × 224 ×3
Epochs	50
Batch Size	32
Activation Function	ReLU
Dropout	0.5
Loss Function	BinaryCrossEntropy
Optimizer	AdamW
Learning Rate	1e-4

C. ABLATION STUDIES

This section aims to examine the effectiveness of the key components integrated within the proposed FoMoSkinNet framework to achieve optimal performance. The evaluation process is designed to investigate the contribution and efficiency of three distinct configurations: (1) the FocalNet branch, (2) the LFNet branch, and (3) the proposed hybrid FoMoSkinNet architecture. To comprehensively assess architectural efficacy, a series of controlled experiments is conducted, analyzing the performance of each individual network as well as the combined hybrid structure.

1) ANALYSIS OF FocalNet

The performance of the FocalNet is evaluated as an independent component to analyze its feature extraction capability in lesion images. The model is designed to effectively encode global structural information and long-range dependencies through its focal modulation mechanism, which replaces the conventional self-attention module. This mechanism enables each token to adaptively aggregate multi-level contextual information, thereby enhancing the network's representational capacity. The training and validation performance of FocalNet is depicted in Figure 10 and Figure 11. The model has attained a 96.88% testing accuracy, a precision of 95.91% and a recall of 97.93% for benign lesions, as shown in Table 6. For malignant lesions, FocalNet has attained a precision

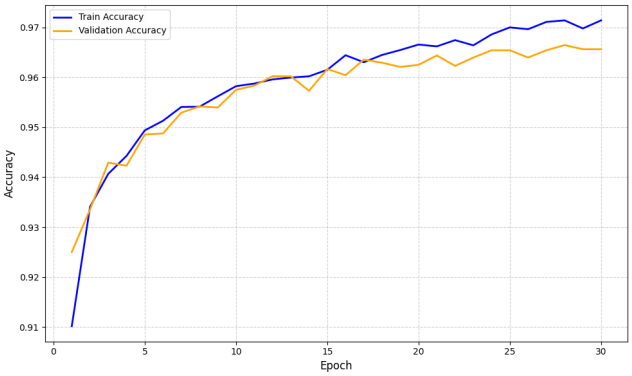


FIGURE 10. Training and validation accuracy of FocalNet.

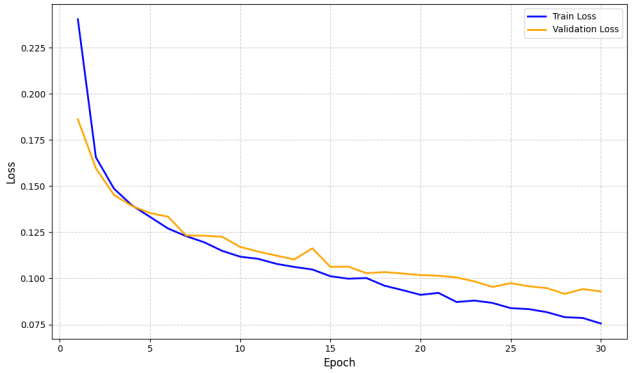


FIGURE 11. Training and validation loss of FocalNet.

TABLE 6. Class-wise evaluation of FocalNet.

Class	Precision (%)	Recall (%)	F1-Score (%)
Benign	95.91	97.93	96.91
Malignant	97.88	95.83	96.85

of 97.88% and a recall of 95.83%, indicating a balanced ability to correctly detect malignant lesions with minimal false positives. The corresponding F1-scores of 96.91% and 96.85% for benign and malignant classes, respectively, confirm the model's consistent and reliable performance across both categories.

2) ANALYSIS OF LFNet

The LFNet serves as another key component of the proposed FoMoSkinNet architecture. It is primarily designed to capture fine-grained spatial and texture-level patterns from lesion images. This network enhances model ability to learn detailed local structures that are critical for accurate lesion classification. The training and validation performance of LFNet is illustrated in Figure 12 and Figure 13. The model has obtained a 90.47% testing accuracy, with class-wise performance summarized in Table 7. Specifically, LFNet has attained a precision, recall, and F1-score of 89.39%,

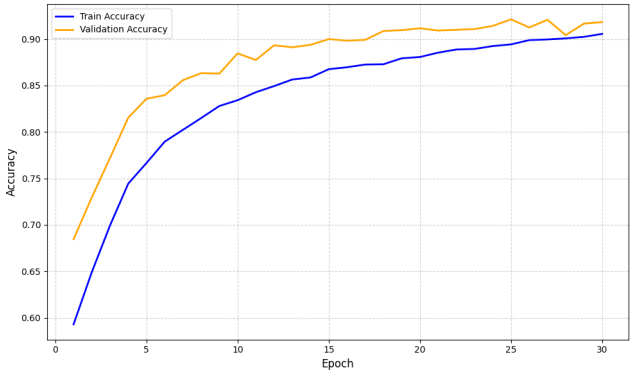


FIGURE 12. Training and validation accuracy of LFNet.

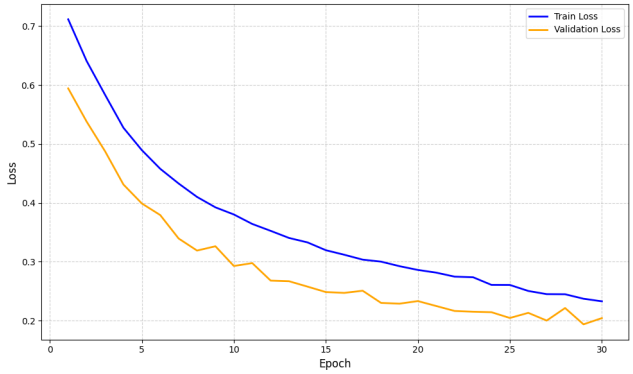


FIGURE 13. Training and validation loss of LFNet.

TABLE 7. Class-wise evaluation of LFNet.

Class	Precision (%)	Recall (%)	F1-Score (%)
Benign	89.39	91.83	90.59
Malignant	91.60	89.10	90.33

91.83%, and 90.59% for Benign class, and 91.60%, 89.10%, and 90.33% for Malignant class. These results demonstrate that LFNet effectively learns localized discriminative features across both lesion classes.

3) ANALYSIS OF THE PROPOSED FoMoSkinNet

The integration of the FocalNet and LFNet tracks is evaluated to examine the combined effect of global and local feature extraction within the proposed FoMoSkinNet framework. The FocalNet branch efficiently captures long-range dependencies and global contextual relationships through focal modulation, while the LFNet branch focuses on extracting fine-grained spatial and texture-level details from lesion images. The fusion of these complementary capabilities enables model to learn more comprehensive and discriminative feature representation, leading to improved lesion classification accuracy. The training and validation performance of FoMoSkinNet is illustrated in Figure 14 and Figure 15.

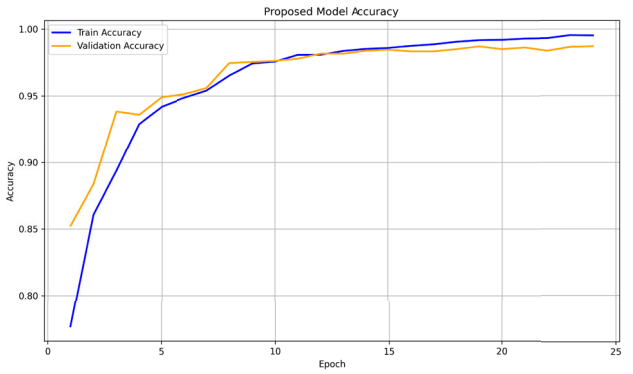


FIGURE 14. Training and validation accuracy of FoMoSkinNet.

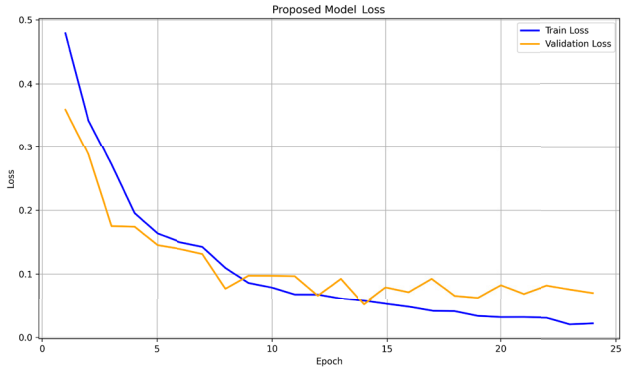


FIGURE 15. Training and validation loss of FoMoSkinNet.

The model has achieved a testing accuracy of 98.85%, with class-wise evaluation metrics summarized in Table 8. For the benign class, FoMoSkinNet has achieved a precision, recall, and F1-score of 98.25%, 99.47%, and 98.86%, respectively. For the malignant class, the corresponding values is 99.46%, 98.23%, and 98.84%. These outstanding results demonstrate ability of the model to capture both fine-grained local textures and global contextual relationships, leading to robust and highly accurate classification.

TABLE 8. Class-wise evaluation of FoMoSkinNet.

Class	Precision (%)	Recall (%)	F1-Score (%)
Benign	98.25	99.47	98.86
Malignant	99.46	98.23	98.84

These ablation studies validate the efficiency of the proposed hybrid design, revealing that each component of FocalNet, LFNet, and their combination within FoMoSkinNet significantly contributes to overall performance improvement. FoMoSkinNet is evaluated with varying numbers of LFNet blocks (three, four, and five). Among these, the four-block configuration has achieved the most balanced performance, offering superior accuracy. The analysis clearly shows that the joint modeling of local and global representations

substantially enhances feature learning and classification accuracy. The results presented in Table 10 summarize comparative performance of all evaluated configurations, emphasizing the efficacy and necessity of the integrated architecture in advancing skin lesion classification tasks.

4) STATISTICAL SIGNIFICANCE ANALYSIS

McNemar's test is performed to evaluate whether the prediction differences between FoMoSkinNet and the other models were statistically significant [47]. This test examines paired nominal data, comparing the number of samples correctly classified by one model but misclassified by another. Specifically, b denotes the number of samples correctly classified by FoMoSkinNet but misclassified by the comparison model, whereas c represents the number of samples misclassified by FoMoSkinNet but correctly classified by the comparison model. The results in Table 9 indicate that FoMoSkinNet significantly outperforms the baseline architectures, with p-values < 0.001 in all pairwise comparisons. The obtained results clearly demonstrate that FoMoSkinNet consistently achieves a substantially higher number of correct predictions compared to all models. In every comparison, the value of b is significantly larger than c, indicating that FoMoSkinNet correctly classifies many samples that competing models misclassify, while making comparatively few additional errors. Consequently, the large McNemar χ^2 statistics and extremely small p-values confirm that these improvements are statistically significant and not due to random variation. These findings validate the robustness and superior generalization capability of the proposed model.

TABLE 9. Statistical analysis of FoMoSkinNet.

rowcolorcyan!30 **Model Comparison**	**b**	**c**	**McNemar χ^2**	**p-value**
FoMoSkinNet vs DenseNet121	1194	22	1127.66	< 0.001
FoMoSkinNet vs ViT	1102	24	1030.13	< 0.001
FoMoSkinNet vs LFNet	528	37	424.95	< 0.001
FoMoSkinNet vs FocalNet	144	38	60.58	< 0.001

D. COMPARISON OF FoMoSkinNet WITH STATE-OF-THE-ART METHODS

To ensure a comprehensive evaluation, we have compared FoMoSkinNet with both modern transformer-based architectures and established CNN baselines. The pretrained transformer models we have used in the experimentation are Vision Transformer (ViT), Swin Transformer and MaxViT. For CNN-based model comparision DenseNet121, InceptionV3, MobileNetV2, ConvNeXt, ResNet50, and VGG16 are used. Recent medical imaging studies continue to use these architectures as strong baselines due to their robustness and reproducibility across datasets. To ensure a similar level of complexity, all the models are reimplemented and trained on the same benchmark ISIC 2024 dataset as in the proposed network under the same experimental conditions, including the same input image size, data augmentations, optimizer settings, learning rate, number of epochs, and dataset split. Without finetuning, we remove the original top classification layers in order to adapt each pretrained architecture to binary skin cancer classification. These are replaced with custom layers tailored to the problem, including a global average pooling layer followed by fully connected layers and a sigmoid activation function for binary output. Table 11 presents comparative results of widely used pretrained architectures. Among the evaluated models, transformer-based networks such as ViT and Swin achieved higher accuracy levels (97% and 96%, respectively) compared to conventional CNN architectures, which attained 75% and 83% accuracy. From Table 11, we can analyze the performance metrics of these models. FoMoSkinNet obtained highest accuracy of 98.85%, with superior values across other evaluation metrics.

E. COMPARISON OF FoMoSkinNet WITH EXISTING STUDIES

A skin lesion classification framework's effectiveness cannot be assessed independently; rather, its full worth is shown when its performance is compared to other diagnostic models. For this reason, a variety of newly published deep learning and transformer-based techniques in the literature are compared with the proposed FoMoSkinNet. It is important to note that such comparisons are non-trivial, as prior studies differ substantially in terms of datasets, class definitions, imaging modalities, and evaluation protocols. To ensure a meaningful comparison, only commonly reported metrics are considered, and preference is given to studies evaluated on public benchmark repositories such as ISIC, HAM10000 and PAD-UFES-20. Since skin cancer diagnosis is highly sensitive to misclassification, metrics reflecting clinical reliability specifically, sensitivity, specificity, and ROC-AUC are prioritized alongside total accuracy. Sensitivity and specificity are particularly critical in skin cancer diagnosis, as they directly reflect a model's ability to correctly identify malignant lesions while minimizing false diagnoses of benign cases. Table 12 summarizes the quantitative performance of FoMoSkinNet in comparison with existing methods. We have considered all aforementioned evaluation measures. The proposed FoMoSkinNet demonstrates superior performance across all key evaluation metrics, highlighting its robustness, reliability, and enhanced capability in skin cancer classification. More importantly, it maintains a strong balance between sensitivity (98.23%) and specificity (99.47%), which is crucial in clinical screening scenarios where both missed malignancies and false alarms must be minimized. These results further confirm the strong generalization ability of the proposed framework in overcoming the limitations of prior approaches, while effectively capturing discriminative and context-aware features from non-dermoscopic images.

TABLE 10. Overall performance comparison of LFNet, FocalNet, and the proposed FoMoSkinNet model.

Model	Accuracy (%)	Specificity (%)	Precision (%)	Recall (%)	F1-Score (%)
LFNet	90.47	91.83	91.60	89.10	90.33
FocalNet	96.88	97.93	97.89	95.93	96.85
FoMoSkinNet (3 Blocks)	98.07	97.37	97.40	98.17	98.08
FoMoSkinNet (4 Blocks)	**98.85**	**99.47**	**99.46**	**98.23**	**98.84**
FoMoSkinNet (5 Blocks)	98.52	98.43	98.44	98.60	98.52

TABLE 11. Comparative evaluation of FoMoSkinNet with Pretrained models.

Model	Parameters (M)	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Avg Inference Time / image
ResNet50	23.85	75	86.21	58.33	69.58	26.83 ms
VGG16	14.78	83	89.12	75.67	81.85	38.14 ms
InceptionV3	22.07	92	92.36	91.13	91.74	14.49 ms
MobileNetV2	2.42	93	98.25	92.17	93.25	11.10 ms
DenseNet121	7.17	95	96.88	94.23	95.54	33.12 ms
ConvNeXt	87.69	67	87.02	40.43	55.21	232.10 ms
ViT	85.90	95	97.00	98.30	97.09	0.22 ms
Swin Transformer	86.87	96	96.96	95.07	96.89	0.80 ms
MaxViT	118	95	96.25	94.00	95.11	6.88 ms
FoMoSkinNet (Proposed)	**29.48**	**98**	**99.46**	**98.23**	**98.84**	**0.18 ms**

TABLE 12. Performance comparison with existing works.

Author	Model	Dataset	Acc	Sen	Spec	Pre	F1	ROC-AUC
[48]	MT-TransUNet	ISIC2017, ISIC, PH2	91.2%	76.93%	94.35%	-	-	92.7%
[28]	Swin-SimAM network	ISIC2017	-	-	-	74%	90%	-
[26]	Mutual Attention Transformer (MAT)	HAM10000	92.55%	-	-	-	-	98.28%
[49]	EnDBoT	ISIC2017	92.1%	90.1%	91.9%	-	94.54%	-
[50]	Assist-Dermo (SqueezeNet-Light)	PH2, ISBI-2017, HAM10000, ISIC	95.6%	94%	96%	94.12%	95.2%	95%
[51]	Innovative 9-layer ViT	ISIC2017	86%	77%	37%	-	-	83%
[52]	DeepMetaForge	ISIC2020	94.1%	88.2%	94.9%	91.2%	91.4%	-
[53]	MelanoTech	SIIM-ISIC	92%	-	-	-	-	-
[54]	Swin + DGSNLA	HAM10000	94.21%	-	-	96.62%	96.64%	-
[55]	DK-Metric Transformer	ISIC2019, PH2, PAD-UFES-20	89.75%	-	-	-	-	84.02%
[56]	xCViT	ISIC Dataset	93.19%	-	92.89%	93.25%	93.19%	96.26%
[57]	Hybrid (MobileNet+ meta)	SLICE-3D ISIC2024	97.7%	99.2	-	94.2%	96.7%	-
[58]	Skin EHDLF	SLICE-3D ISIC2024	98.76%	98.3%	99.0%	99.2%	98.3%	98.7%
Our	**FoMo SkinNet**	**SLICE-3D ISIC2024**	**98.85%**	**98.23%**	**99.47%**	**99.46%**	**98.84%**	**99.84**

Overall, FoMoSkinNet successfully integrates the complementary strengths of convolutional and transformer-based architectures, resulting in highly accurate and consistent classification outcomes across all benchmark metrics.

F. CROSS-DATASET EVALUATION

To validate the generalization of FoMoSkinNet, we performed cross-dataset evaluation using the PAD-UFES-20 skin lesion dataset [55]. This dataset originally contains six classes of skin lesions. For binary classification, we grouped basal cell carcinoma (BCC), melanoma (MEL), and squamous cell carcinoma (SCC) as malignant, and actinic keratosis (ACK), melanocytic nevus (NEV), and seborrheic keratosis (SEK) as benign. All images are preprocessed in the same manner as the ISIC 2024 dataset, including resizing and normalization. FoMoSkinNet has achieved robust performance across all evaluation metrics, as shown in Figure 16. Table 13 presents the class-wise performance of FoMoSkinNet on the cross-dataset PAD-UFES-20 evaluation. These results highlight the effectiveness of the FoMoSkinNet in capturing both local and global features, enabling accurate lesion classification on unseen datasets.

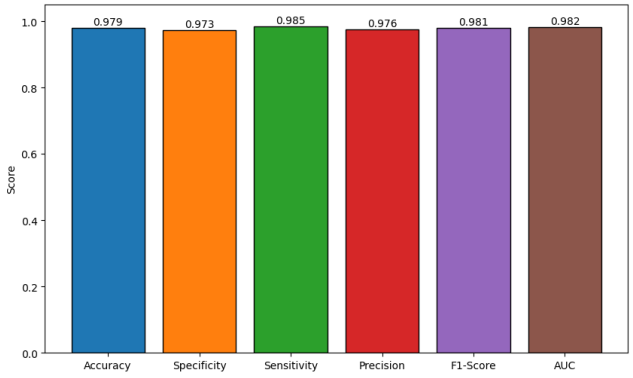


FIGURE 16. Performance of FoMoSkinNet on PAD-UFES-20 dataset.

TABLE 13. Class-wise evaluation of FoMoSkinNet on PAD-UFES-20 dataset.

Class	Precision (%)	Recall (%)	F1-Score (%)
Benign	98.33	97.25	97.79
Malignant	97.59	98.54	98.06

VII. CONCLUSION

Skin cancer remains a significant global health challenge, where early and accurate diagnosis plays a vital role in ensuring effective treatment and improved survival rates. This study proposed FoMoSkinNet, a DL-based approach designed to automatically classify lesions as benign or malignant using non-dermoscopic images. The proposed model leverages a dual-track architecture combining FocalNet for global feature extraction and LFNet for local feature learning, enabling a more comprehensive feature representation. Experimental evaluations demonstrated that FoMoSkinNet achieved superior performance compared to existing techniques, achieving an F1-score of 98.84%, accuracy of 98.85%, precision of 99.46%, and sensitivity of 98.23%. Evaluation against several SOTA models further confirmed its reliability and effectiveness. While FoMoSkinNet demonstrates strong performance, it has been trained and evaluated on non-dermoscopic images. Future work could extend its application to other imaging modalities. Additionally, integrating explainable AI techniques will make the model's predictions more interpretable and trustworthy for medical professionals.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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